National University of Computer and Emerging Sciences
Lahore Campus

Digital Logic Design (EE1005)

Final Exam

Date: May 23, 2026

Total Time (Hrs.): 03
Total Marks: 90
Total Questions: 5

Course Instructor(s)
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Instruction/Notes:
- *Answer all the questions in the space provided. **NO answer booklet is required.***
- *Show proper working for each question to get credit*
- *The answer to each question should be readable.*
- You may use a rough sheet, but **do not attach it** as it will not be graded
- ***Use of calculator is not allowed.***

CLO # 1: Understand different number systems and their conversion.

Question#1: (a): Without going into divisions and multiplications, convert the following binary number into hexadecimal, octal and base-4. ***Show working for each part.*** ***[Marks: 3+6+6]***

$(1\ 0\ 1\ 1\ 1\ 0\ 1\ 0\ 0\ 0\ 1\ 0\ 1\,.\,1\ 0\ 1\ 1\ 1)_2$

i) 1 7 4 5 . B 1 $)_{16}$

ii) 1 3 5 0 5 . 5 6 $)_8$

iii) 1 1 3 1 0 1 1 . 2 3 ~~1~~ 2 $)_4$

3

R. Work
1 7 4 5 .
1 0111 0100 (101
01

(b): ASCII is a 7-bit alpha-numeric code. There are 128 codes from 0000000 to 1111111. The codes of these alpha-numeric characters are normally summarized in a table with 16 rows and 8 columns. The rows are designated by $B_4B_3B_2B_1$ that range from 0000 to 1111. The columns are designated by $B_7B_6B_5$ that range from 000 to 111.
If the ASCII code for the letter "R" is 1010010, then find out;
i) The ASCII code for the letter "U"?, and
ii) The ASCII code for the letter "M"?

1 3 5 0 5 .
1 011 101 000 (
101

1 1 3 1 0 1 1 . 2 3 1
01 01 11 01 00 01 01 . 10 11 1

The ASCII code for the letter "U" is 1010101.

The ASCII code for the letter "M" is 1001101

(e): Without converting into any other base, perform direct multiplication of the two base-5 numbers given below, to obtain a product in base-5.

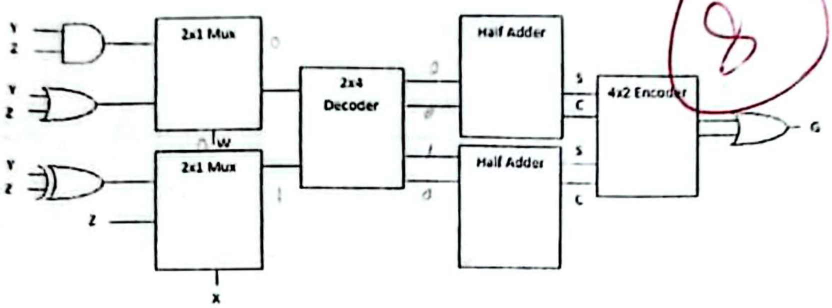
$$
\begin{array}{rrrrrrr}
 & & & (3 & 1 & 4 & 2)_5 \\
 & & \times & (4 & 3 & 2)_5 \\
\hline
 & & 1 & 1 & 3 & 3 & 4 \\
1 & 2 & 0 & 0 & 3 & 1 & \times \\
2 & 3 & 2 & 2 & 3 & \times & \times \\
\hline
(3 & 0 & 3 & 4 & 4 & 4 & 4)_5
\end{array}
$$

6

CLO # 2: Construct optimized combinational circuit diagram.

Question#2 (a): For the circuit given below find value for the output G at following input tuples: *[Marks: 10+5]*

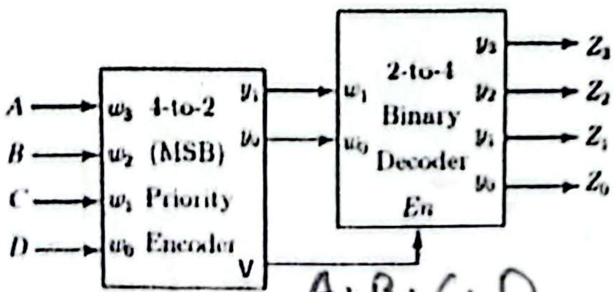
8



W	X	Y	Z	G
0	0	0	1	1
0	0	1	1	0
1	0	0	0	0
1	1	0	1	1
1	1	1	0	0

: Determine Boolean expressions for the outputs from Z_3 through Z_0 in terms of inputs A through

Note: *The priority encoder shown in the logic circuit given above has w3 the highest priority till the lowest priority input.*



A+B+C+D

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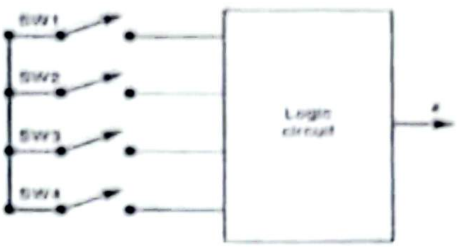
$Z_3 = w_1 w_0$
$Z_2 = w_1 \overline{w_0}$
$Z_1 = \overline{w_1} w_0$
$Z_0 = \overline{w_1}\,\overline{w_0}$

zero

$Y = A + B + C + D$

CLO#2: Construct optimized combinational circuit diagram.

Question#3 (a): SW1, SW2, SW3, SW4 are switches that are part of the control circuitry in a copy machine. The switches are at various points along the path of the copy paper as the paper passes through the machine. Each switch is normally open (binary 0), and as the paper passes over a switch, the switch closes (binary 1). Design the logic circuit to detect and produce a HIGH (output $X = 1$) whenever two or more switches are closed at the same time. Also, switches SW1 and SW4 should not be closed at the same time. ***[Marks: (5+5)+(5+5)]***



Requirements:

1. Construct a truth table clearly indicating all possible input combinations and corresponding output
2. Find simplified expression for X using K-Map

SW1	SW2	SW3	SW4	X
0	0	0	0	0
0	0	0	1	0
0	0	~~0~~ 1	0	0
0	0	~~0~~ 1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
	0	0	0	0
	0	0	1	0
	0	1	0	0
	0	1	1	0
	1	0	0	
	1	0	1	
	1	1	0	
	1	1	1	

4

SW1 SW2 \ SW3 SW4	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	x	x	1
10	0	x	x	1

Expression:

$= \overline{SW1}\,SW2\,SW4 + SW1\,SW2\,\overline{SW3}$
$SW2\,SW3 + SW1\,SW3$

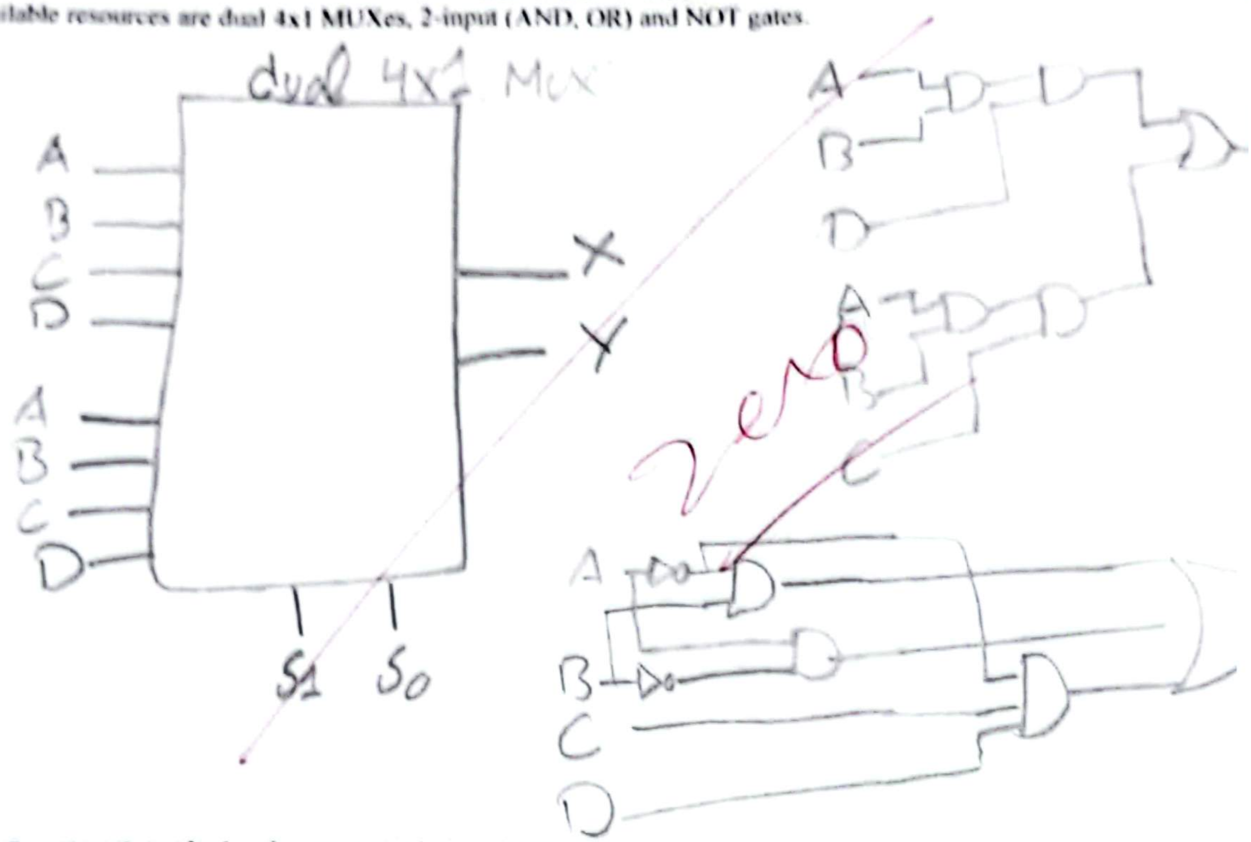
1

CamScanner

(b): Design a device which has four inputs and two outputs using MUX's and minimum extra gates (if needed). Where

$$X = ABD + ABC$$
$$Y = AB + AB + ACD$$

Available resources are dual 4x1 MUXes, 2-input (AND, OR) and NOT gates.



CLO#3: Construct optimized sequential circuit diagram.

<u>Question#4</u> (a): A vending machine is controlled by a Finite State Machine (FSM). The machine operates using four 2-bit input codes: Coin Inserted = 00, Drink Button Pressed = 01, Payment Confirmed = 10, and Drink Ready = 11. The FSM follows the sequence: 00 → 01 → 10 → 11 ***[Marks: 10+10]***

The FSM transitions to the next state only when the correct next input code is received. The FSM behavior is defined as follows:

1. If the correct next input is received, the FSM moves to the next state.
2. The FSM does not allow sequence skipping. If an input corresponding to a non-adjacent next state is received, the FSM remains in the current state until the correct sequence input is applied.
3. If the input corresponding to a previous state is received, the FSM returns to that previous state, allowing the customer to modify the previous operation before final completion.
4. Once the drink reaches the Drink Ready state, if the Coin Inserted input (00) is repeated, the FSM remains in the same state to complete the current order and does not move backward.
5. At the final stage, the drink is ready and the FSM moves back towards the initial state.

Draw **State diagram** of vending machine FSM ***ONLY.***

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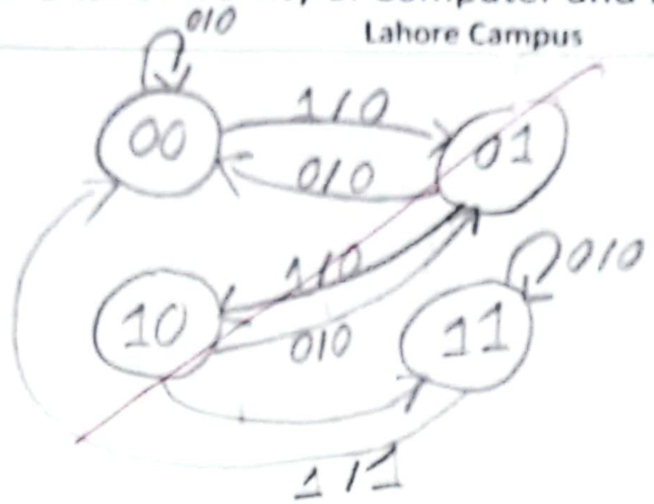
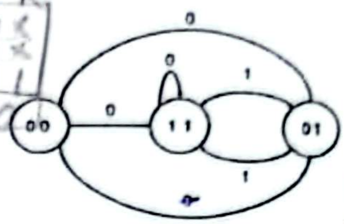


zero

(b): For the state diagram given fill the following ***state table*** utilizing JK Flip Flops (we don't care about un-used states)

X	Q1(t)	Q0 (t)	Q1(t+1)	Q0(t+1)	J1	K1	J0	K0
0	0	0	1	1	1	X	1	X
0	0	1	0	0	0	X	X	1
0	1	0	X	X	x	x	X	X
0	1	1	1	1	X	0	X	0
1	0	0	0	1	0	X	1	X
1	0	1	1	1	1	X	x	0
1	1	0	X	X	x	X	X	X
1	1	1	d	1	X	1	x	0

5

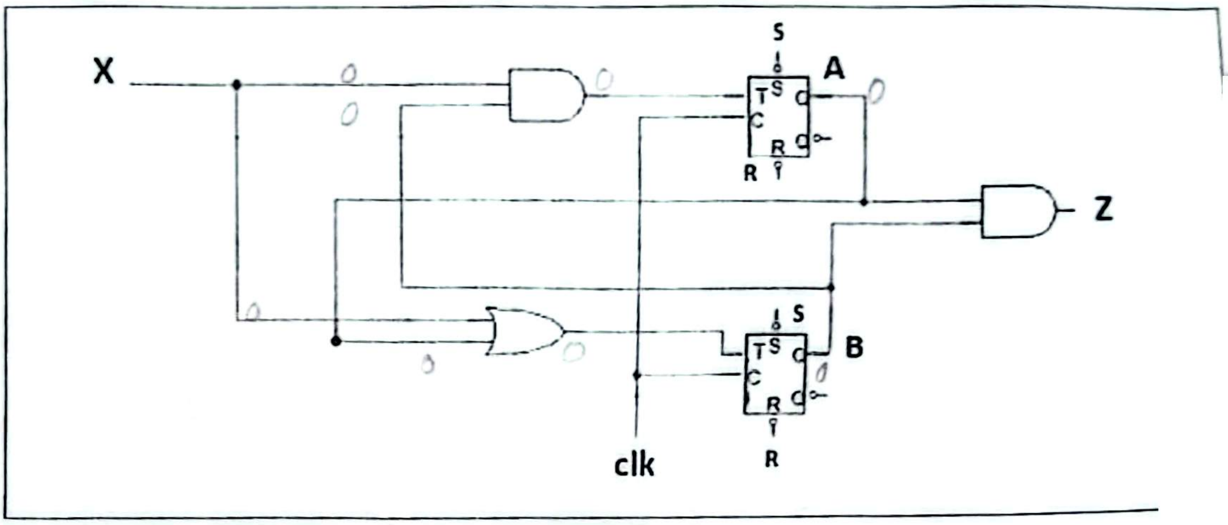


Filp Flop Input Equations:

2 ?

? cw

Question#5 (a): Fill in the state table for the sequential circuit given below: ***[Marks: 10+10]***



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State Table:

4+1.5

X	A(t)	B(t)	A(t+1)	B(t+1)	Z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	0	0
1	1	0	1	0	0
1	1	1	0	0	1

(b): A research lab is designing a **4-bit register system** to handle different control commands. The system uses **two selection inputs (S1, S0)** to decide which operation will be performed on the register contents. The required operations are:

1. **Data Logging (S1S0 = 00):** Whenever new sensor readings arrive, they must be stored directly into the register.
2. **Overflow Simulation (S1S0 = 01):** The register must transform its current contents into a special form that represents what would happen if the value were doubled, but only the lower 4 bits are kept. **Note:** ***DONOT USE*** ***ARITHMETIC MULTIPLICATION OR ADDITION METHOD*** for this operation.
3. **Code Verification (S1S0 = 10):** The register contents are masked with the fixed security code 1010 to selectively activate only the relevant bit positions, while all other bits remain inactive.
4. **Parity Check (S1S0 = 11):** The register must be processed such that the parity checker generates output indicating whether the number of 1's in 4 bits content is even or odd, and this output should be displayed on all outputs of the register (1111 for odd and 0000 for even).

Draw the complete circuit diagram showing the 4-bit register, selection inputs, control logic. **Use the available resource list only: (D flip flops, 2-input (AND, OR, XOR, XNOR, NAND, NOR) gates, NOT gates, Multiplexers).** Clearly indicate how each selection input activates the correct operation